Mathematics - II (For CE, CSE, ME) (January- June 2026)
Tutorial Sheet - II (Partial Differential Equations)

(1) Form the PDE by eliminating arbitrary constants from

(i)$z^2(1+a^3) = 8(x+ay+b)^3$ (iii)$z = ae^{bx}\sin by$

(ii)$z = axe^y + \frac{1}{2}a^2e^{2y} + b$ (iv) $log(az-1) = x+ay+b$

(2) Form the PDE by eliminating arbitrary functions from

(i)$z = f(xy/z)$ $(iii) f(x+y+z, x^2+y^2+z^2) = 0$

(ii)$z = \phi(x+y)\psi(x-y)$ (iv)$z = f(x) + e^y g(x)$

(3) Obtain general integral of the following PDE

(i) $p\sqrt{x} + q\sqrt{y} = \sqrt{z}$ (iii)$p+3q = 5z + tan(y-3x)$

(ii)$p-q = log(x+y)$ (iv) $p\cos(x+y) + q\sin(x+y) = z$

(4) Find the integral surface of the PDE

$x(y^2+z)p - y(x^2+z)q = (x^2-y^2)z$

containing the straight line $x+y=0, z=1$.

(5) Find a complete solution of the following partial differential equations

(i)$3p^2 - 2q^2 = 4pq$ (ii) $p(1-q^2) = q(1-z)$

(iii)$z = px + qy + 3(pq)^3$ (iv) $p+q = \sin x + \sin y$

(v)$(x-y)(px-qy) = (p-q)^2$ (vi)$4xyz = pq + 2px^2y + 2qxy^2$
(vii) $pe^y = qe^x$

(viii)$p^2z + q^2 = (\frac{2}{3})^2$ (ix)$q^2y^2 = z(z-px)$ (x) $\sqrt{p} + \sqrt{q} = 1$

(6) Solve the following partial differential equations using Charpit's method

(i)$pxy + pq + qy = yz$ (ii)$z = p^2x + q^2y$(iii)$\frac{z}{px} = \frac{qy}{z}$

(iv)$16p^2z^2 + 9q^2z^2 + 4z^2 - 4 = 0(v)2(xp + yq + z) = yp^2$

(vi)$p^2+qy = -2(z+y^2)$(vii)$z = \frac{p}{q}(1 + p^2)+b; b$ being a constant

(7) Find the general solution of the following partial differential equations

(i)$(D^4 - D^3D' + 2D^2D'^2 - 5DD'^3 + 3D'^4)z = 0$

(ii)$(D^3 - 4D^2D' + 5DD'^2 - 2D'^3)z = e^{2x+y}$

(iii)$(D^2 + 3DD' + 2D'^2)z = 24xy$

(iv)$(D^2 - DD' - 2D'^2)z = (y - 1)e^x$

(v)$(D^3 - 3DD'^2 + 2D'^3)z = \sqrt{x + 2y}$

(vi)$(D^2 - DD')z = \cos 2y(\sin x + \cos x)$

(vii)$(4D^2 - 4DD' + D'^2)z = 16log(x + 2y)$

(viii)$(D^3 + D^2D' - DD'^2 - D'^3)z = e^x cos(2x + 3y)$

(8) Find the general solution of the following partial differential equations

(i)$(2D + D' + 5)(D - 2D' + 1)^2z = 0$

(ii)$(D^2 - D'^2 + 3D - 3D')z = 0$

(iii)$(D^2 + DD' + D' - 1)z = \sin(x + 2y)$

(iv)$(D^2 - DD' + +D')z = x^2 + y^2$

(v) $(D - 3D' - 2)^2z = 2e^{2x} \tan(y + 3x)$

(vi)$\big((D + D')^3 - 4(D + D')^2 + 3(D + D')\big)z = e^{x+y+2} \cos(2x - y)$.

(9) Solve (by method of separation of variables:)

$(i)(D^2 - 2D + D^{'})u = 0.$

$(ii)Du = 2D^{'}u + u,$ given that $u(x, 0) = 10e^{-5x}.$

$(iii)py^3 + qx^2 = 0.$